PERFECTNESS DESCENDS ALONG FPQC COVERINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Derived pullback preserves perfect complexes. invoked at proof lines 3125-3126. Proves the only-if direction.
- Perfect complexes have quasi-coherent cohomology sheaves, via the pseudo-coherent lemma of this chapter and the cohomology perfect lemma. invoked at proof lines 3128-3131. Starts the converse: each pullback has quasi-coherent cohomology sheaves.
- For flat morphisms the cohomology sheaves of the derived pullback are the pullbacks of the cohomology sheaves. used at proof line 3132. Transfers the quasi-coherence question to E itself.
- fpqc descent of quasi-coherence, by the Descent chapter proposition. invoked at proof lines 3133-3134. Concludes the cohomology sheaves of E are quasi-coherent.
- The cohomology perfect lemma characterizes perfect complexes among objects with quasi-coherent cohomology by bounded tor-amplitude and pseudo-coherence, both of which descend along fpqc coverings by two lemmas of this chapter. invoked at proof lines 3135-3139. Assembles the converse direction.

Source reference. Original lemma 09UA: sources/stacks-chow/tags.txt, source lines 12418–12418. Cited at proof lines 3125-3126 for the only-if direction; the converse is inside the counted span.

Source reference. Original lemma 08E5: sources/stacks-perfect/pinned-original-source.tex, source lines 2422–2422. Cited at proof lines 3128-3129 for quasi-coherence of the cohomology sheaves of a perfect object; the descent conclusion is inside the counted span.

Source reference. Original lemma 08CQ: sources/stacks-chow/tags.txt, source lines 10610–10610. Cited at proof lines 3130-3131 and again 3135-3137; the assembly of the converse direction is inside the counted span.

Source reference. Original proposition 023T: sources/stacks-chow/tags.txt, source lines 2754–2754. Cited at proof lines 3133-3134 to descend quasi-coherence of the cohomology sheaves; the conclusion is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09UD: sources/stacks-perfect/pinned-original-source.tex, source lines 3024–3024. Cited at proof lines 3138-3139 in the final assembly; the conclusion is inside the counted span.

Source reference. Original lemma 09UE: sources/stacks-perfect/pinned-original-source.tex, source lines 3045–3045. Cited at proof lines 3138-3139 in the final assembly; the conclusion is inside the counted span.

Lemma 0.2. *Let $\{f_i : X_i \to X\}$ be an fpqc covering of schemes. Let $E \in D(\mathcal{O}_X)$. Then E is perfect if and only if each Lf_i^*E is perfect.*

Proof. Pullback always preserves perfect complexes, see Cohomology, Lemma 09UA. Conversely, assume that Lf_i^*E is perfect for all i. Then the cohomology sheaves of each Lf_i^*E are quasi-coherent, see Lemma 08E5 and Cohomology, Lemma 08CQ. Since the morphisms f_i is flat we see that $H^p(Lf_i^*E) = f_i^*H^p(E)$. Thus the cohomology sheaves of E are quasi-coherent by Descent, Proposition 023T. Having said this the lemma follows formally from Cohomology, Lemma 08CQ and Lemmas 09UD and 09UE. □